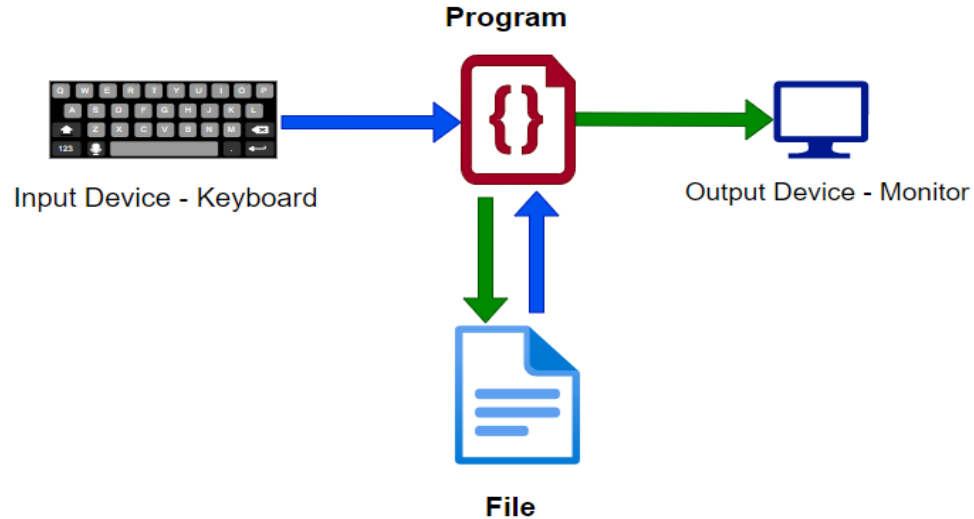


How to Read and Write to a File in C++



Example 1: Open and Close a File

In this example program, we will demonstrate how to open/create a file and how to close the file in C++.

```
#include <fstream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    ofstream myFile_Handler;
```

```
    // File Open
```

```
    myFile_Handler.open("File_1.txt");
```

```
    // File Close
```

```
    myFile_Handler.close();
```

```
    return 0;
```

```
}
```

Example 2: Write to a File

We can write to a file using the stream insertion operator, i.e., "<<". In this program, we have used the file handler and insertion operator to write two lines in the file. The insertion operator ("<<") indicates that we are inserting the string into the output file stream object.

```
#include <fstream>
using namespace std;
int main()
{
    ofstream myFile_Handler;
    // File Open
    myFile_Handler.open("File_1.txt");

    // Write to the file
    myFile_Handler << "This is a sample test File. " << endl;
    myFile_Handler << "This is the second line of the file. " << endl;

    // File Close
    myFile_Handler.close();
    return 0;
}
```

Example 3: Read from a File

In the previous examples, we showed you how to write content to a file. Now, let's read the content from the file that we created in Example-2 and display the content on the standard output device, i.e., the monitor.

We use the `getline()` function to read the complete line from the file and then "cout" to print the line on the monitor.

```
#include <iostream>
#include <fstream>
#include <string>

using namespace std;

int main()
{
    ifstream myFile_Handler;
    string myLine;

    // File Open in the Read Mode
    myFile_Handler.open("File_1.txt");

    if(myFile_Handler.is_open())
    {
        // Keep reading the file
        while(getline(myFile_Handler, myLine))
        {
            // print the line on the standard output
            cout << myLine << endl;
        }
        // File Close
        myFile_Handler.close();
    }
    else
    {
        cout << "Unable to open the file!";
    }
    return 0;
}
```

Example 4: Read and Write to

To both read and write to a file, we get an fstream object and open it in “ios::in” and “ios::out” mode.

In this example, we first write some data to the file. Then, we read the data from the file and print it to the monitor.

```
#include <string>
using namespace std;

int main()
{
    fstream myFile_Handler;
    string myLine;

    // File Open
    myFile_Handler.open("File_1.txt", ios::in | ios::out);

    // Check if the file has opened
    if(!myFile_Handler.is_open())
    {
        cout << "File did not open!";
        exit(1);
    }

    // Write to the file
    myFile_Handler << "1. This is another sample test File. " << endl;
    myFile_Handler << "2. This is the second line of the file. " << endl;

    myFile_Handler.seekg(ios::beg);

    // Read the File
    if(myFile_Handler.is_open())
    {
        // Keep reading the file
        while( getline(myFile_Handler, myLine))
        {
            // print the line on the standard output
            cout << myLine << endl;
        }

        // File Close
        myFile_Handler.close();
    }
    else
    {
    }
```